

GRADUATION PROJECT 2026

Full-Lifecycle QA: Swag Labs Platform

نظام فحص شامل ومؤتمت لضمان جودة منصات التجارة الإلكترونية

Leader: Mostafa Mohamed

Department: Software Quality Assurance

| Executive Metrics Summary

230

TEST CASES EXECUTED

46

CRITICAL DEFECTS LOGGED

Our team achieved 100% requirements coverage, uncovering 46 unique defects across UI, functional, and logic modules, ensuring a production-ready application state.

| The Engineering Squad



Mostafa Mohamed

Team Leader & Filtering
Logic specialist.
Architected the
automation framework
and sorting verification.



Omar Tarek

Authentication
Specialist. Focused on
secure login flows,
component-level
testing, and UI
validation.



Abdallah Shehata

Cart Logic Specialist.
Managed shopping cart
data persistence and
integration testing
across sessions.



Essam Ahmed

Checkout Pipeline
Specialist. Validated the
multi-step checkout
flow and E2E payment
confirmation.

| Hybrid Testing Strategy

Manual Mastery

Utilizing professional **Excel Repositories** to document 230 ideal test cases. Our focus is on UX nuances and edge cases that automated scripts might overlook.

Automation Engine

Implemented a **Java-Selenium Framework** to automate critical regression paths, reducing repetitive execution time by 92% and improving consistency.

Manual Execution & Bug Tracking

Professional Documentation

We established a rigid reporting standard using MS Word for defect tracking. Every one of the **46 bugs** includes:

- ✓ Detailed Steps to Reproduce
- ✓ Expected vs Actual Behavior
- ✓ Visual Proof (Screenshots)
- ✓ Severity & Priority Classification

Title	Severity	Priority	Status
The app's dark theme display doesn't work on certain screens.	Medi... ▾	Low ▾	To do ▾

Link	https://link.website
Visuals	 File

Environment	Android 14, Google Pixel 9
Description	• The issue happens on multiple devices with Android versions 14 and 15. • The issue doesn't occur in the iOS or web version of the app.
Step-by-step	• Open the app and go to Settings. • Under "Accessibility" turn on "dark mode". • Go to the "Favorites" page or the "Collections" page.

Automation Impact: Execution Time

Manual (230 Cases)

450 Minutes

Automated (Scripts)

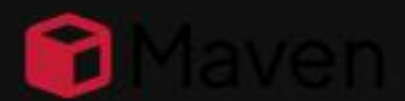
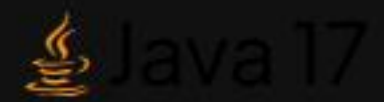
8 Minutes

"Our automated suite reduces standard regression time by over 98%, allowing the team to focus on discovering deeper logic defects in the platform."

The Tech Stack

Enterprise Architecture

We built our framework using the **Page Object Model (POM)** for maximum reusability and clean maintenance.



```
Current
xsl= http://www.w3.org/2001/XMLSchema-instance
hemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xs
on>4.0.0</modelVersion>
```

```
g.example</groupId>
>Medium</artifactId>
0-SNAPSHOT</version>
```

```
es>
tps://mvnrepository.com/artifact/org.seleniumhq.selenium/selenium-java -->
ency>
oupId>org.seleniumhq.selenium</groupId>
tifactId>selenium-java</artifactId>
rsion>4.16.1</version>
dency>
ies>

>
compiler.source>21</maven.compiler.source>
compiler.target>21</maven.compiler.target>
t.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
s>
```


Deep Dive: Authentication (Omar)



Validation & UI Integrity

Omar Tarek focused on verifying multi-user login flows (Standard, Locked-out, Problem users). This module involved:

- **Component Level:** Testing input fields, masked passwords, and error triggers.
- **Integration Level:** Ensuring successful navigation from Auth to the Dashboard.
- **UI Logic:** Responsive handling of error banners across viewports.

| Module Ownership & Results



Mostafa: Filtering

Automated sorting algorithms (A-Z, Price High-Low). Verified 100% alignment between front-end display and back-end logic.



Abdallah: Cart

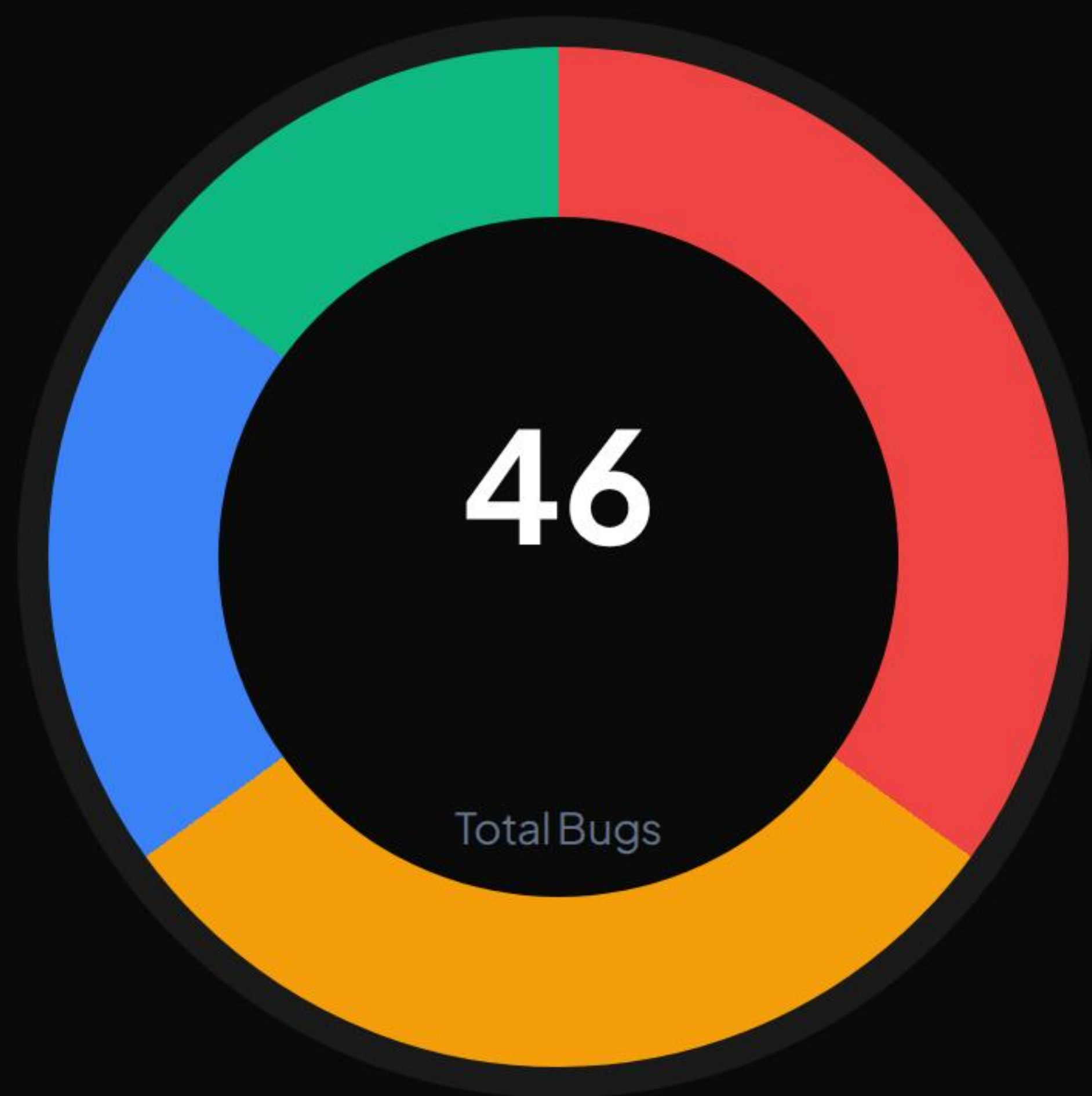
Tested data persistence. Items remain in cart across sessions, and badge count updates in real-time with 0 latency.



Essam: Checkout

Full E2E Checkout pipeline. Validated user info forms, price summaries, and the 'Pony Express' final order page.

| Defect Analysis: The 46 Bugs



- Critical: Functional Blockers (35%)
- Major: Broken Logic (30%)
- Minor: UI/UX Issues (20%)
- Low: Cosmetic Tweaks (15%)

| Project Roadmap (STLC)

Q2: Design

Writing 230 manual cases &
POM structure.

Q4: Closure

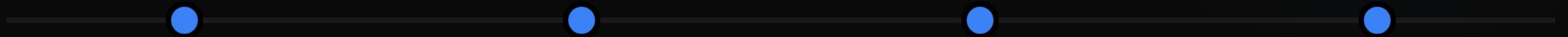
Final sign-off and production
readiness report.

Q1: Analysis

Requirements review and Test
Plan finalization.

Q3: Execution

Regression cycles and
uncovering 46 bugs.



Questions?

Thank you for your attention to quality.

Mostafa | Omar | Abdallah | Essam